

SOAP: Credit Risk Model Redevelopment

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Executive Decision & Performance Transformation

The rebuilt logistic model materially improves applicant ranking over the supplied legacy benchmark. The legacy benchmark showed weak rank discrimination (validation Gini of **0.110**, AUC of **0.555**), which was insufficiently discriminating for the proposed banking use without redevelopment and validation. On common validation data, the rebuilt Generalized Linear Model (GLM) increased Gini by **+0.286 points** to **0.396**, while held-out testing supported similar performance on comparable historical applications, with Gini of **0.374** and AUC of **0.687**.

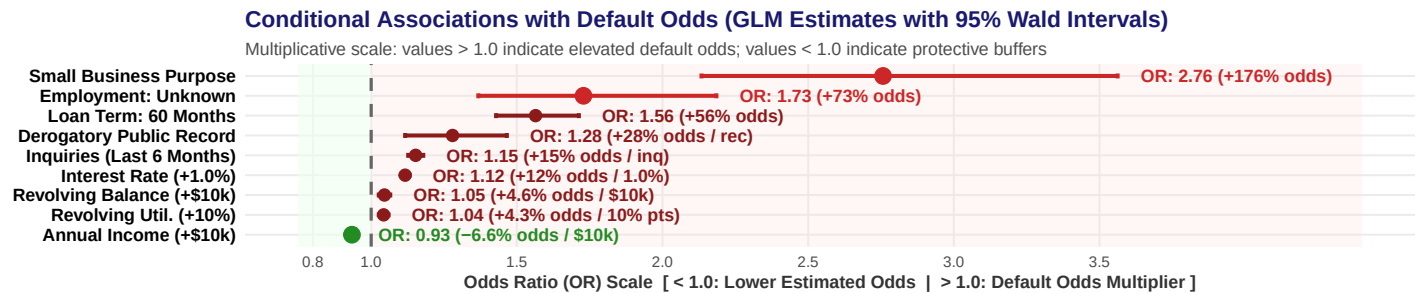
Management Decision: Retain the transparent GLM as the candidate applicant-ranking model, subject to redevelopment or recalibration on contemporary Australian data and independent validation. Reserve the spatial-temporal Generalized Linear Mixed Model (GLMM) for portfolio monitoring rather than live applicant scoring.

Methods: Binomial logistic regression, backward AIC selection, stratified 5-fold cross-validation, held-out validation and testing, ROC/Gini and calibration assessment, and a state-month random-intercept GLMM with forward temporal validation.

Performance Metric	Legacy Benchmark	Rebuilt Model	Operational Takeaway
Validation Discrimination (Gini)	0.110 (AUC = 0.555)	0.396 (AUC = 0.698)	+0.286 validation Gini points on common evaluation sample
Held-Out Test Sample	Not available	Gini 0.374; AUC 0.687	95% CI: AUC [0.670, 0.703]; performance remained moderate on untouched historical test sample
Maximum Separation (KS)	Not available	KS = 0.280 (at 40.1% volume)	Maximum separation between cumulative default and non-default score distributions at 40.1% volume
Illustrative Policy Scenario (15% Cutoff)	Not applicable	60.6% approval volume	Approved default rate: 9.2% (vs 14.9% baseline); captures 62.5% of defaulting accounts

Primary Risk Drivers: Conditional Associations with Default Odds

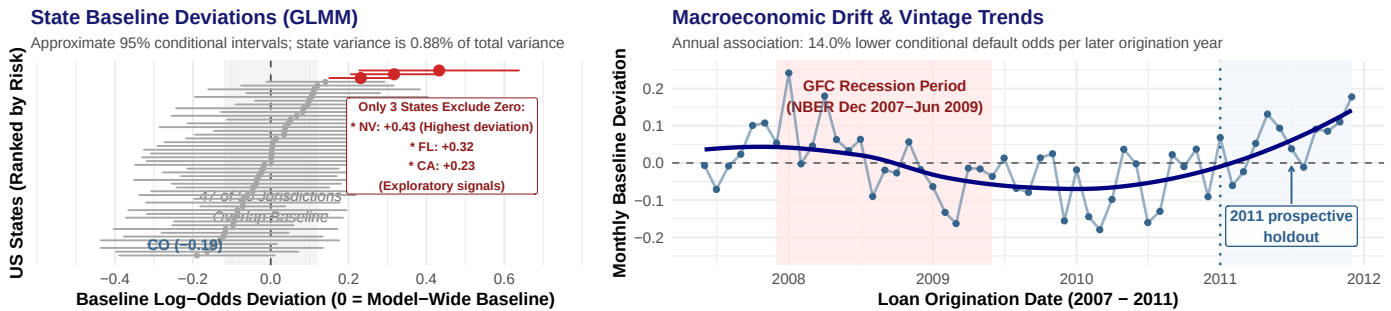
To support explainability, the scorecard estimates conditional Odds Ratios (OR) with 95% Wald confidence intervals. Factors above 1.0 are associated with higher conditional default odds; factors below 1.0 are associated with lower conditional default odds.



- Largest Positive Associations:** Small-business loans carry +175.7% higher estimated conditional default odds (OR = 2.76, 95% CI: [2.13, 3.56]) relative to car loans. Applications with unreported employment tenure have +72.9% higher odds (OR = 1.73, [1.37, 2.19]), and 60-month terms carry +56.4% higher odds (OR = 1.56, [1.43, 1.71]). *Recommendation:* Review small-business-purpose applications under a dedicated policy assessment; treat missing employment tenure as a data-quality signal before operational use.
- Additional Risk Indicators:** Interest rate, inquiries, derogatory public records, revolving balance, and line utilization were all positively associated with default odds. Each additional derogatory public record was associated with 27.9% higher estimated conditional default odds (OR = 1.28, [1.12, 1.47]).
- Lower Estimated Odds:** Higher annual income was consistently associated with lower conditional default odds: OR = 0.934 per additional \$10,000, 95% CI: [0.923, 0.945].
- Expected-but-Dropped Account Depth Variables:** Total accounts (total_acc) and open accounts (open_acc) were not retained by backward AIC after adjustment for other candidate predictors, indicating limited incremental contribution under the selected specification. *Governance note:* Do not remove these fields operationally without local validation and assessment of their broader underwriting uses.

Geographic and Vintage Dynamics: The “Ranking vs. Monitoring” Paradox

State and month random effects improve portfolio description and vintage monitoring, but do not materially improve applicant ranking:



- **The “Ranking vs. Monitoring” Finding:** On the 2011 temporal holdout, adding the fitted state and time effects did not materially improve applicant rank discrimination: GLM Gini **0.4070** versus GLMM Gini **0.4069**. Borrower-level characteristics therefore provided essentially all of the rank-ordering performance achieved by the evaluated models on this holdout.
- **Recommended Operational Role:** Advance the parsimonious GLM as the candidate applicant-ranking architecture for local redevelopment, recalibration, and independent validation. Use the GLMM for historical vintage monitoring, geographic portfolio decomposition, and identifying cohorts requiring investigation.
- **Fair Lending & Governance Guardrails:** Nevada, Florida, and California had positive estimated baseline deviations with approximate conditional intervals excluding zero (state variance represents 0.88% of latent variance). These exploratory historical associations do not establish causality. **Management must not implement regional rejection quotas or geographic risk pricing**, which could create material fairness, legal, and governance concerns.
- **Temporal Trend Association:** Holding observed borrower characteristics constant, later origination years were associated with lower default odds (OR = 0.860, 95% CI: [0.798, 0.927]). Possible contributors include underwriting changes, borrower composition, outcome seasoning, and broader historical conditions.

Diagnostic Findings, Governance Limitations & Deployment Roadmap

To support model-governance review consistent with credit-risk management expectations, we evaluated model diagnostics and operational requirements:

- **Diagnostic Findings:**
 - *Global Checks:* Overall residual uniformity ($p = 0.811$) and dispersion ($p = 0.756$) tests were non-significant; adjusted GVIF values below 1.6 indicate no severe multicollinearity likely to destabilize fitted main-effect estimates.
 - *Qualifications:* Grouped risk-band dispersion ($p = 0.020$) identified uneven variability across coarse predicted-risk groups, with mild underprediction in several intermediate-to-high risk bands; monthly aggregate residuals retained positive serial dependence ($DW = 0.984$, $p < 0.001$), indicating persistence between adjacent cohorts.
- **Prudential Roadmap for Australian Banking Deployment:**
 - *Representative Australian Data:* Before deployment, redevelop or recalibrate the model using representative Australian application, repayment, credit-bureau, affordability, and macroeconomic data, followed by independent validation and fairness assessment.
 - *Complete Expected Credit Loss Framework:* The model provides Probability of Default (PD) only. Capital and expected-loss applications require validated PD, Loss Given Default (LGD), and Exposure at Default (EAD) components ($ECL = PD \times LGD \times EAD$) under approved governance.